COMPETENCY STANDARD

FOR

REFRIGERATION AND AIR CONDITIONING

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity.
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-RAC-01-G	Perform Computations Using Basic Mathematical Formulas	1. Identify calculation requirements in the workplace 2. Identify and select appropriate mathematical methods. 3. Use tool/instrument to perform calculations	10
SICIP-LE-RAC-02-G	Communicate in English in the Workplace	1. Read, understand and prepare documents in English 2. Prepare workplace documents in English 3. Perform conversations in English	10
SICIP-LE-RAC-03-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	10
		Total Hour	**30**

Sector Specific Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-RAC-01-S	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-RAC-02-S	Work with Hand Tools, Power Tools and Electrical Testing Instruments	1. Inspect hand tools and power tools for usability 2. Use hand tools, electrical testing instruments properly and safely 3. Operate power tools properly and safely 4. Clean/maintain hand tools and power tools after use	12
SICIP-LE-RAC-03-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 1. Implement waste management practices	08
		Total Hour	**30**

Occupation Specific Competencies (300 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-RAC-01-O	Understand Basic Electrical System	1. Interpret fundamental principles of electricity and electronics 2. Carry out basic electrical circuit 3. Perform cables joints 4. Understand different types of electrical loads and load calculation 5. Use electrical device, components and other associated elements 6. Maintain and store electrical/electronic tools/instruments	60
SICIP-LE-RAC-02-O	Perform Tube Processing Operation	1. Carry out tube processing 2. Cut tubes 3. Flare tube ends 4. Swage tube end 5. Bend tube 6. Braze tubes 7. Clean/maintain workplace, tools and equipment	30
SICIP-LE-RAC-03-O	Service and Maintain Refrigerators and Freezers	1. Carry out Preparation for servicing and maintenance works 2. Troubleshoot refrigerator/freezer 3. Maintain and repair refrigerator/freezer 4. Service mechanical refrigeration system 5. Clean/maintain workplace, tools and equipment	40
SICIP-LE-RAC-04-O	Service and Maintain Window Type Air Conditioning System	1. Carry out preparation for servicing and maintenance works 2. Troubleshoot window air conditioner 3. Maintain/repair electrical /electronic system 4. Service mechanical refrigeration system	30

Code	Unit of Competency	Elements of Competency	Duration (hours)
		5. Install window type air conditioner 6. Clean/maintain workplace, tools and equipment	
SICIP-LE-RAC-05-O	Service and Maintain Split and Package Type Air Conditioning system	1. Carry out preparation for servicing and maintenance works 2. Troubleshoot split and package type air conditioning units 3. Maintain/repair electrical and electronic system 4. Service mechanical system and components 5. Install split type air conditioner 6. Clean/maintain workplace, tools and equipment	120
SICIP-LE-RAC-06-O	Carry out Compressor Servicing	1. Carry out preparation for compressor servicing/repairing 2. Diagnose fault and service compressor 3. Clean/maintain workplace, tools and equipment	20
		Total Hours	**300**

COMPETENCY STANDARD: REFRIGERATION AND AIR CONDITIONING

The Generic Competencies

Unit of Competency: **PERFORM COMPUTATIONS USING BASIC MATHEMATICAL FORMULAS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-RAC-01-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical concepts in workplace. It specifically includes the tasks of identifying calculation requirements in the workplace, identifying and selecting appropriate mathematical methods, and using appropriate tools and instruments to perform calculations.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace	1.1 **Calculation requirements** are identified 1.2 **Workplace information** are interpreted
2. Identify and select appropriate mathematical methods	2.1 **Appropriate calculation method** is identified 2.2 Appropriate calculation method is selected
3. Use tool/instrument and perform calculations	3.1 Appropriate **tools and instruments** are identified 3.2 Computation is made

Range of variables:

Variable	**Range** May include but not limited to:
1. Calculation requirements.	1.1 Area 1.2 Volume 1.3 Measurements: Length/Width/Height/thickness/diameter 1.4 Weight and density Calculation
2. Workplace information	2.1 Drawing 2.2 Part drawing 2.3 Verbal instructions 2.4 Job order and Job sheet
3. Appropriate calculation method	3.1 Addition, Subtraction, Division and Multiplication 3.2 Conversion of measurement 3.3 Percentage and ratio calculation 3.4 Geometric Formulas (Square, Rectangle, Circle, Sphere etc)
4. Tools/instruments	4.1 Calculator 4.2 Computer

Curricular Content Guide

1. Underpinning Knowledge	1.1 Numerical concept, identification of calculation requirements 1.2 Basic mathematical methods such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas. 1.3 Mathematical language, symbols. 1.4 Measuring units
2. Underpinning Skills	2.1 Computing basic mathematics. 2.2 Practicing conversion of measurements 2.3 Calculating areas, volume etc using formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect to peers, sub-ordinates and seniors in workplace 3.5 Environmental concern 3.6 Sincerity and honesty
4. Resource Implications	The following resources must be provided. 4.1 Stationeries 4.2 Consumables 4.3 Calculators 4.4 Computers 4.5 Measuring tape

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified calculation requirements from workplace information. 1.2 Selected appropriate method to carry out the calculation requirements. 1.3 Completed calculations using appropriate tools/instruments.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: COMMUNICATE IN ENGLISH IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-RAC-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to communicate in English in the workplace. It specifically includes the tasks of reading, understanding and preparing documents in English, preparing workplace documents in English and performing conversations in English.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Read, understand and prepare documents in English	1.1 Workplace documents are read and understood 1.2 Visual information is interpreted. 1.3 Resume/CV is understood and prepared
2. Prepare workplace documents in English	2.1 Simple **routine workplace documents** are prepared using key words, phrases, simple sentences 2.2 **Visual aids** are prepared and submitted. 2.3 Key information is written in the appropriate places in standard forms.
3. Perform conversations in English	4.1 Active listening is demonstrated. 4.2 Conversation is performed in English with peers, customers and management as per the workplace standard.

Range of Variables

Variable	**Range** May Include but not limited to:
1. Routine workplace documents	1.1 Meeting Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs
2. Visual aids	2.1 Maps 2.2 Diagrams 2.3 Forms 2.4 Labels 2.5 Graphs 2.6 Charts

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Interpret workplace documents in English 1.2 Preparing workplace documents in English 1.3 Listen to conversation in English 1.4 Job roles, responsibilities and compliances 1.5 Understanding of visual aids (e.g., graphs, presentations, or charts) and their preparation 1.6 Knowledge of workplace communication etiquette (e.g., formal vs. informal conversations). 1.7 Awareness of workplace standards for professional communication.
2. Underpinning Skills	2.1 Reading, Understanding, and Preparing Documents in English 2.2 Preparing Workplace Documents in English 2.3 Performing Conversations in English Language 2.4 Interpersonal skills to maintain professional communication in workplace settings
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect of peers, sub-ordinates and seniors in workplace 3.5 Environmental concern 3.6 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Work place Procedure 4.2 Materials relevant to the proposed activity 4.3 All tools, equipment, material and documentation required. 4.4 Relevant specifications or work instructions

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Read, Understood, and Prepared Documents in English 1.2 Conversed in English with peers and customers. 1.3 Made reports of workplace documents in English.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A SELF-DIRECTED TEAM**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-RAC-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal **forms of communication** are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member.	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practice. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range May Include but not limited to:

1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Sharing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice. 2.8 Team member's support to other members is appreciated and valued Identifying current and potential problems faced by the team 2.9 Identifying solutions to the problem 2.10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Teamwork

	3.2 Promptness in carrying out activities. 3.3 Tidiness and timeliness. 3.4 Respect of peers, sub-ordinates and seniors in workplace. 3.5 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Work books 4.5 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

A. The Sector Specific Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-RAC-01-S
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **OHS policies** and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 **Personal Protective Equipment (PPE)** is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 **Hazards and risks** are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 **Emergency response plans and procedures** are implemented. 4.3 **First aid procedure** is applied during emergency situations.

Range of Variables

Variable	**Range** May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes

	2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness

	3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Practiced personal health and safety procedures 1.4 Reported incidents arising from hazards and risks to authority. 1.5 Implemented plans and procedures to respond emergency 1.6 Applied basic first aid procedure.
4. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
5. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK WITH HAND TOOLS, POWER TOOLS AND ELECTRICAL TESTING INSTRUMENTS**	**Nominal Duration:** 12 hrs.	**Unit Code:** SICIP-LE-RAC-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work with hand tools and power tools properly. It specifically includes the tasks of inspecting hand tools, power tools and electrical testing instruments for usability, using hand tools and electrical testing instruments properly and safely, operating power tools properly and safely and cleaning/maintaining hand tools and power tools after use.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hand tools, power tools and electrical testing instruments for usability	1.1 Appropriate tools are selected. 1.2 Application of tools to job requirement is determined. 1.3 Usability of tools are checked and verified. 1.4 **<u>Hand tools and electrical testing instruments/measuring devices</u>** are prepared. 1.5 Sources of power supply for power tools identified.
2. Use hand tools and electrical testing instruments properly and safely	2.1 Appropriate hand tools, electrical testing instruments for the job are used. 2.2 Proper and safe use/operation is applied in the different types of hand tools. 2.3 **<u>Safety precautions</u>** is observed when using hand tools. 2.4 Unsafe or faulty tools are identified and marked for repairing
3. Operate power tools properly and safely	3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.2 Proper sequence of operation is applied in using power tools to produce results. 3.3 Power tools are used safely in accordance to manufacturer's operating specification.
4. Clean/maintain hand tools and power tools after use	4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard. 4.2 Scraps or wastages are removed and cleaned 4.3 Condition of tools is checked after use. 4.4 Appropriate lubricant is applied after use and prior to storage. 4.5 Measuring tools are checked and calibrated. 4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced.

Range of Variables

Variable	Range May include but not limited to:
1. Hand tools and electrical testing instruments /measuring devices	1.1 Electrical tools 1.1.1 Side cutter pliers 1.1.2 Set of screw drivers 1.1.3 Set of combination wrenches 1.1.4 Adjustable wrench 1.1.5 Wire stripper 1.1.6 Electrician's knife 1.1.7 Measuring tape 1.1.8 Electrical Hand Drill 1.2 Electrical testing Instruments 1.2.1 Multi-meter/AVO meter 1.2.2 Clamp meter 1.2.3 Volt meter 1.2.4 Ammeter 1.2.5 Power meter 1.2.6 Megger tester 1.2.7 Soldering iron with stand. 1.2.8 Solder sucker. 1.2.9 Neon tester. 1.3 Mechanical Tools 1.3.1 Cutting pliers. 1.3.2 Nose pliers. 1.3.3 Combination pliers. 1.3.4 Measuring tape. 1.3.5 Tweezers. 1.3.6 Scissors. 1.3.7 Wire striper 1.3.8 Power Saw 1.3.9 Drill Machine
2. Safety precautions	2.1 Use of appropriate PPEs 2.2 Proper hand, feet and eye coordination 2.3 Working environment 2.4 Safe operating condition of hand tools and power tools 2.5 Awareness to OHS requirements

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of tools, functions and use 1.2 Types of Hand tools and their proper use and techniques 1.3 Types of electrical testing instruments and their functions 1.4 Types of Power tools, use and safe handling method 1.5 Technical application of tools 1.6 Procedures in the use of hand tools and power tools

	1.7 Policies and procedures for occupational health and safety 1.8 Reporting and documentation 1.9 Preventive maintenance 1.10 Methods and techniques 1.11 Quality procedures 1.12 Waste management 1.13 Storage procedures
2. Underpinning Skills	2.1 Using appropriate hand tools for the job 2.2 Observing safety precautions when using hand tools 2.3 Using electrical testing instruments or devices 2.4 Using power tools correctly and safely in accordance to manufacturer's operating specification. 2.5 Checking condition of tools after use 2.6 Applying appropriate lubricant on hand tools and power tools after use and prior to storage 2.7 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories 2.8 Storing Tools, electrical testing instruments/devices and power tools safely in appropriate location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Environmental concerns 3.3 Eagerness to learn 3.4 Tidiness and timeliness 3.5 Concern to proper use of tools 3.6 Orderliness
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Operation and maintenance manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Used appropriate hand tools for the job. 1.2 Observed safety precautions when using hand tools. 1.3 Used power tools safely in accordance to manufacturer's operating specification. 1.4 Cleaned and maintained hand tools and power tools after use and prior to storage. 1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories.

	1.6 Stored tools and power tools safely in appropriate location.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES**	**Nominal Duration:** 08 hrs.	**Unit Code:** SICIP-LE-RAC-04-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Principles of green practices</u>** are explained. 1.2 **<u>Sources of environmental impacts</u>** during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 **<u>Ways of mitigating environmental impacts</u>** in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 **<u>Recyclable and non-recyclable items</u>** are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 **<u>Different types of waste</u>** are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	**Range** May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental impact

	1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Light Engineering sector

	1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings 1.3 Sources of environmental impacts in Light Engineering 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in Light Engineering activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to sustainable and eco-friendly Light Engineering practices 3.2 Active participation in reducing resource use and waste generation 3.3 Eagerness to adopt green habits in professional life 3.4 Promptness in implementing sustainable practices in Light Engineering activities 3.5 Tidiness and concern for environmental cleanliness 3.6 Compliance with safety standards and environmental regulations
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waster segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in Light Engineering sector 1.2 Identified sources of environmental impacts

	1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: **UNDERSTAND BASIC ELECTRICAL SYSTEM**	**Nominal Duration:** 60 hrs.	**Unit Code:** SICIP-LE-RAC-01-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to understand basic electrical system. It specifically includes the tasks of interpreting fundamental principles of electricity and electronics, carrying out basic electrical circuit, performing cables joints, understanding different types of electrical loads and load calculation, using electrical device, components and other associated elements and maintaining and storing electrical/electronic tools/instruments		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	**Performance Criteria**
1. Interpret fundamental principles of electricity and electronics	1.1 **<u>Fundamental principles</u>** of electricity and electronics are described. 1.2 Application of OHM's law is explained. 1.3 **<u>Electrical/Electronic properties/parameters</u>** are measured using appropriate measuring tool/instrument or comprehended
2. Carry out basic electrical circuit	2.1 **<u>Basic elements</u>** for preparing electrical circuits are identified and collected 2.2 Soldering is performed and practiced. 2.3 **<u>Electrical/electronic circuit diagram</u>** is understood and analyzed. 2.4 Basic electrical circuits wiring are carried out.
3. Perform cables joints	3.1 Different types of **<u>cables joints</u>** used in industry are explained 3.2 Characteristics and size of the cables are interpreted. **3.3** Cable joints are carried out as workplace standard
4. Understand different types of electrical loads and load calculation	4.1 **<u>Resistive loads</u>** are identified and explained 4.2 **<u>Inductive loads</u>** are identified and explained 4.3 **<u>Capacitive loads</u>** are identified and explained
5. Use electrical device, components and other associated elements	5.1 **<u>Electrical device, components and other associated elements</u>** are identified, used and interpreted. 5.2 **<u>Electronic components</u>** are used safely/properly.
6. Maintain and Store electrical/electronic tools/instruments	6.1 Electrical/ electronic tools/instruments are checked for proper operation. 6.2 Electrical/ electronic tools/instruments are maintained in accordance to manufacturer's specification. 6.3 Electrical/ electronic tools/instruments stored in accordance to workplace procedures/policy.

Range of Variables

Variable	**Range** (Includes but not limited to:)
1. Fundamental principles	1.1 Ohms Law 1.2 Kirchhoff's law 1.3 Principles and theory of AC/DC circuits 1.4 Series and parallel circuits 1.5 Law of conductivity 1.6 Law of resistivity 1.7 Semiconductor 1.8 Half wave and full wave rectification
2. Electrical/electronic properties/parameters	2.1 Voltage 2.2 Resistance 2.3 Current 2.4 Impedance 2.5 Capacitance 2.6 Resistance 2.7 Inductance 2.8 Reactance
3. Basic elements	3.1 Switch 3.2 Socket 3.3 Cable 3.4 Resistive load 3.5 Inductive load
4. Electrical/electronic circuit diagram	4.1 Series circuit 4.2 Parallel circuit 4.3 Series-parallel circuit 4.4 Lamp circuits 4.5 Refrigerator circuit 4.6 Window air conditioner circuit 4.7 Deep freezer circuit
5. Basic electrical circuit	5.1 Series circuits 5.2 Parallel circuits 5.3 Series-parallel circuits 5.4 One Lamp controlled from one point 5.5 One Lamp controlled from two point 5.6 Two Lamps controlled from two point, 5.7 Two Lamps controlled from three point 5.8 Tube Light Connection 5.9 Power supply circuit
6. Cable joints	6.1 Straight Through Joint 6.2 Branch Joint 6.3 Tee Joint 6.4 Tap Joint 6.5 Soldered Joint
7 Resistive loads	7.1 Electric bulb 7.2 Heater coil

	7.3 Electric Iron 7.4 Rice Cooker
8 Inductive loads	8.1 Induction motor 8.2 Electric fan 8.3 Magnetic Conductor 8.4 Electric choke coil/Ballast
9 Capacitive loads	9.1 Capacitor load bank 9.2 PFI plant
10.Electrical device, components and other associated elements	10.1 Power supply 10.2 Compressor motor 10.3 Relay 10.4 Capacitor 10.5 Timer 10.6 Thermostat/Cooling overload sensor 10.7 Switches 10.8 Fuse 10.9 Contactor 10.10 Lamps 10.11 Terminal block
11Electronic components	11.1 Resistor. 11.2 Diode. 11.3 Capacitor. 11.4 Inductor. 11.5 Transistor (P-type and N-type) 11.6 Thyristors and IC.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental principal of electricity 1.2 Definition of AC and DC current and their use in circuits 1.3 Types of DC circuits and their application 1.4 AC circuits and their application 1.5 Differential electrical/electronic circuit diagrams applied in refrigeration and air conditioning 1.6 Circuit wiring, installation and maintenance 1.7 Cable joints and different types of joints 1.8 Resistive, inductive and capacitive loads and their uses in different circuits 1.9 Functions of electrical devices and electronic components and applications 1.10 Electrical measurement and testing methods and techniques 1.11 Safety precautions when working with electrical circuits 1.12 Basic wiring circuits and their application 1.13 Electrical lighting systems
2. Underpinning Skills	2.1 Carrying out basic electrical circuit diagramming and wiring

	2.2 Describing relationships of the different types of electrical properties 2.3 Measuring electrical parameters using appropriate measuring tool/instrument 2.4 Using electrical/electronic measuring tools and testing instruments safely/properly 2.5 Preparing basic electrical circuits 2.6 Testing power supply and electrical/electronic components in accordance with manufacturer's specifications 2.7 Terminating electrical/electronic circuit components in accordance with given diagram 2.8 Testing circuit for proper operation in accordance with work instruction/circuit design 2.9 Storing electrical tools/instruments in accordance to workplace procedures/policy
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Concern to environmental care 3.3 Eagerness to learn 3.4 Tidiness, timeliness, and orderliness 3.5 Respect for rights of peers and seniors in workplace 3.6 Communication with peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Various kinds of electrical/electronic components, tools and equipment 4.3 Workplace rules and regulation policy manual 4.4 Pens 4.5 Papers 4.6 Work books

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Carried out basic electrical circuit diagramming and wiring 1.2 Performed different types of cable joints 1.3 Measured electrical parameters using appropriate measuring tool/instrument 1.4 Used resistive, inductive and capacitive loads in different electrical/electronic circuits and performed testing with instruments safely/properly 1.5 Tested power supply and electrical components in accordance 1.6 Terminated electrical/electronic circuit components in accordance with given diagram 1.7 Tested circuit for proper operation in accordance with work instruction/circuit design 1.8 Stored electrical tools/instruments in accordance to workplace procedures/policy

2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM TUBE PROCESSING OPERATION	Nominal Duration: 30 hrs.	Unit Code: SICIP-LE-RAC-02-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform tube processing operation. It specifically includes the tasks of carrying out tube processing, cutting tubes, flaring tube ends, swaging tube end, bending tube, brazing tubes and cleaning/maintaining workplace, tools and equipment		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out tube processing	1.1 **<u>Tools, equipment & materials</u>** are gathered. 1.2 Tools, equipment & materials are checked for usability and operability. 1.3 **<u>Tube dimensions</u>** and **<u>fittings</u>** are identified and gathered. 1.4 Measurements and pipe runs are identified in accordance with workplace requirements/specifications.
2. Cut tubes	2.1 **<u>Tubes</u>** are measured and marked in accordance with specification. 2.2 Tubes are cut using by appropriate cutting method and tool. 2.3 Tubes are reamed on its ends after cutting to remove burrs. 2.4 Tube ends are sealed to ensure non contamination with dirt and **<u>foreign materials.</u>** 2.5 Appropriate **<u>sealing material</u>** is used on tube ends.
3. Flare tube end	3.1 Tube ends are flared using appropriate flaring tool. 3.2 Flared tube end is checked for quality. 3.3 Flared tube ends are sealed to ensure non-contamination with dirt and foreign materials.
4. Swage tube end	4.1 Tube ends are swaged using appropriate swaging tool. 4.2 Swaged tube end is checked for quality. 4.3 Swaged tube end is sealed to ensure non-contamination with dirt and foreign materials.
5. Bend tube	5.1 Tube is bended using appropriate bending tool. 5.2 Bended copper/aluminum tube is checked for quality in accordance with specifications 5.3 Bended copper/aluminum tubes are sealed to ensure non- contamination with dirt and foreign materials.
6. Braze tubes	6.1 **<u>Brazing equipment</u>** is checked for usability and safety condition. 6.2 Tubes are brazed using appropriate brazing equipment 6.3 Brazed joints are checked for quality.

	6.4 Brazed connection is tested in accordance with workplace requirements/specification.
7. Clean/maintain workplace, tools and equipment	7.1 Workplace is cleaned and materials are stored in accordance with workplace requirements. 7.2 Tools and equipment are cleaned, checked for damaged and lubricated (if necessary) and stored in accordance with workplace conditions. 7.3 Damaged/defective tools and equipment are reported for repair/replacement

Range of Variables

Variable	**Range** (Includes but not limited to:)
1. Tools, equipment & materials	1.1 Tools; 1.1.1 Measuring steel tape 1.1.2 Hammer 1.1.3 Tube cutter 1.1.4 Hand hacksaw 1.1.5 Swaging tool set 1.1.6 Flaring tool set 1.1.7 Bending tool 1.1.8 Files 1.1.9 Bench vice 1.2 **E**quipment 1.2.1 Oxy-acetylene welding set 1.2.2 Drill press 1.3 Materials 1.3.1 Copper tubes 1.3.2 Aluminum tubes 1.3.3 Brazing flux 1.3.4 Cotton rag 1.3.5 Brazing filler materials 1.3.6 Emery Paper
2. Tube dimensions	2.1 Tube size (1/4, ½, ¾, 5/8, 1) in inch 2.2 Length 2.3 Radius/diameter 2.4 Angle of bend
3. Fittings	3.1 Flare nut 3.2 Coupling 3.3 Elbow 3.4 Tube plug/cap 3.5 Union 3.6 Lock Kit
4. Tubes	4.1 Copper tube 4.2 Aluminum tube
5. Foreign materials	5.1 Water 5.2 Sand 5.3 Dust

	5.4 Metal filings 5.5 Copper filings 5.6 Aluminum filings
6. Sealing material	6.1 Plastic caps 6.2 Rubber caps 6.3 Tapes 6.4 Tube plug 6.5 Tube caps
7. Brazing equipment	7.1 Oxy-acetylene welding set 7.2 LPG gas welding set 7.3 Blow torch

Curricular Content Guide

1. Underpinning Knowledge	1.1 Tube cutting tools and their application 1.2 Tube cutting procedure and techniques 1.3 Types of tubes seals and their use 1.4 Tube flaring tools and their application 1.5 Tube swaging, bending and brazing tubes 1.6 Procedure of tube flaring 1.7 Method of checking flared tube quality 1.8 Swaging tool and its application 1.9 Procedure of swaging tube 1.10 Types of tube benders and application 1.11 Procedure and technique of bending copper/aluminum tubes 1.12 Types of equipment used for brazing and their application 1.13 Copper tube brazing procedure and techniques 1.14 Aluminum tube brazing procedure and technique 1.15 Determining brazed joint quality 1.16 Procedure of testing brazed connection
2. Underpinning Skills	2.1 Cutting tubes using appropriate cutting method and tool 2.2 Sealing tube ends to ensure non contamination with dirt and foreign materials 2.3 Flaring tube ends using appropriate flaring tool 2.4 Checking flared tube end for quality 2.5 Swaging tube end using appropriate swaging tool 2.6 Bending copper/aluminum tube using appropriate bending tool 2.7 Brazing copper tubes using appropriate brazing equipment 2.8 Brazing aluminum tubes using appropriate brazing equipment 2.9 Checking brazed joints for quality 2.10 Testing brazed connection in accordance with workplace requirements/specification.
3. Underpinning Attitudes	3.1 Patience

	3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns 3.4 Eagerness to learn 3.5 Tidiness and orderliness 3.6 Respect to peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Refrigeration and air conditioning equipment, tools, materials, repair and training manual. 4.3 Work instruction sheet 4.4 Copper, aluminum and steel tube

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Cut tubes using appropriate cutting method and tool 1.2 Sealed tube ends to ensure non contamination with dirt and foreign materials 1.3 Flared tube ends using appropriate flaring tool 1.4 Checked flared tube end for quality 1.5 Swaged tube end using appropriate swaging tool 1.6 Bended copper/aluminum tube using appropriate bending tool 1.7 Brazed copper and aluminum tubes using appropriate brazing equipment 1.8 Tested brazed connection in accordance with workplace requirements/specification. 1.9 Cleaned workplace and stored materials, tools and equipment in accordance with workplace requirements
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE AND MAINTAIN REFRIGERATORS AND FREEZERS**	**Nominal Duration: 40 hrs**	**Unit Code:** SICIP-LE-RAC-03-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to service and maintain refrigerators and freezers. It specifically includes the tasks of carrying out preparation for servicing and maintenance works; troubleshooting refrigerator/freezer; maintaining and repairing refrigerator/freezer; servicing mechanical refrigeration system; cleaning/maintaining workplace, tools and equipment		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out preparation for servicing and maintenance works	1.1 **Safe work practices** are maintained and Personal Protective Equipment (PPEs) are used. 1.2 Workplace area is inspected and hazards are eliminated 1.3 **Tools, equipment & materials** are gathered. 1.4 Tools, equipment & materials are checked for usability and quality. 1.5 Workplace is prepared for servicing activities. 1.6 Domestic Refrigerator/freezer is inspected and corresponding **technical information** is identified and interpreted.
2. Troubleshoot refrigerator/freezer	2.1 **Relevant information** regarding trouble/problem is asked from user/owner of unit. 2.2 Electrical wiring circuit is checked and traced. 2.3 Refrigerator/freezer is started and operated, if possible, and observed operation. 2.4 **Electrical, electronic and mechanical parameters** are observed and recorded. 2.5 **System trouble/problem** is identified and results/findings are recorded. 2.6 Electronic soldering circuit is checked and traced.
3. Maintain and repair refrigerator/freezer	3.1 **Refrigerant** is recovered, leaked tested and vacuumed 3.2 Electrical/electronic trouble/problem is identified. 3.3 Inverter Circuit is checked and repaired/replaced. 3.4 Faulty Electrical & electronic components are tested and repaired/replaced where necessary. 3.5 Specifications of electrical/electronic component for replacement is checked and recorded. 3.6 **Electrical/electronic Maintenance activities** are carried out in accordance with **manufacturer's instructions/specification.** 3.7 Operation of electrical components and system is checked and tested for proper operation in accordance with manufacturer's specification.

4. Service mechanical refrigeration system	4.1 **<u>Compressor operation faults</u>** are identified. 4.2 Refrigerant compressor fault is repaired and tested for normal operation. 4.3 **<u>Mechanical refrigeration system fault</u>** is identified and repaired/serviced in accordance with manufacturer's instructions/specifications. 4.4 **<u>Servicing procedures</u>** of refrigeration system components are carried out in accordance with manufacturer's instructions/specifications. 4.5 **<u>Refrigerator/freezer</u>** is tested for acceptable operating performance in accordance with manufacturer's specifications.
5. Clean/maintain workplace, tools and equipment	5.1 Workplace is cleaned and materials are stored in accordance with workplace requirements. 5.2 Tools and equipment are cleaned, checked for damaged and lubricated (if necessary) and stored in accordance with workplace conditions. 5.3 Damaged/defective tools and equipment are reported for repair/replacement.

Range of Variables

Variable	**Range** (Includes but not limited to):	
1. Safe work practices	1.1 Proper and safe use of tools and equipment 1.2 Wearing of appropriate PPEs 1.3 Eliminating unsafe conditions in the workplace 1.4 Shutting off and locking electrical switches before start of work 1.5 Care in working with electricity 1.6 Awareness to working with high pressures 1.7 Awareness in working with refrigerant gasses/liquids Releasing gas pressures before opening high pressure lines	
2. Tools, equipment & materials	2.6 Tools; 2.6.1 Multi tester (AVO) 2.6.2 Clamp meter 2.6.3 Side cutting pliers 2.6.4 Wire stripper 2.6.5 Screwdriver set 2.6.6 Terminal crimping plier 2.6.7 Measuring steel tape 2.6.8 Ball peen hammer 2.6.9 Tube cutter	2.7.3 Refrigerant recovery machine 2.7.4 Oxy acetylene welding outfit 2.7.5 Double Gauge Manifold 2.7.6 Drill press 2.7.7 Welding booth 2.7.8 Nitrogen Cylinder 2.7.9 Double stage Regulator 2.8 Materials 2.8.1 Insulation tapes

<table>
<tr><td></td><td>2.6.10 Hand hacksaw
2.6.11 Swaging tool set
2.6.12 Flaring tool set
2.6.13 Bending tool
2.6.14 Files
2.7 Equipment
2.7.1 Vacuum pump
2.7.2 Refrigerant charging cylinder</td><td>2.8.2 Electrical terminal connectors/clip
2.8.3 Wires/cables
2.8.4 Copper tubes
2.8.5 Aluminum tubes
2.8.6 Steel tube
2.8.7 Brazing flux
2.8.8 Brazing Rod
2.8.9 Cotton rag</td></tr>
<tr><td>3. Technical information</td><td colspan="2">3.1 Refrigerant recovered, vacuumed the system and tested
3.2 Type of refrigerant used,
3.3 High side systems pressure
3.4 Low side system pressure
3.5 Type of compressor used
3.6 Voltage of compressor motor
3.7 Full load current rating
3.8 Electrical/ circuit diagram
3.9 Electrical components
3.10 Type of evaporator coil
Type of condenser</td></tr>
<tr><td>4. Refrigerant</td><td colspan="2">4.1 R-134a
4.2 R-600a
4.3 R-404
4.4 R-290
4.5 CFC, HCFC, HFC and HCs</td></tr>
<tr><td>5. Technical information</td><td colspan="2">Refrigerant recovered, vacuumed the system and tested
5.1 Type of refrigerant used,
5.2 High side systems pressure
5.3 Low side system pressure
5.4 Type of compressor used
5.5 Voltage of compressor motor
5.6 Full load current rating
5.7 Electrical/ circuit diagram
5.8 Electrical components
5.9 Type of evaporator coil
5.10 Type of condenser</td></tr>
<tr><td>6. Relevant information</td><td colspan="2">6.1 Power supply
6.2 Electrical/electronic circuit
6.3 System operation
6.4 Compressor
6.5 Evaporator
6.6 Condenser
6.7 Expansion valve
6.8 Refrigerant charge
6.9 Leaks
6.10 Incidents prior to occurrence of problem</td></tr>
</table>

7. Electrical, electronic and mechanical parameters	Electrical parameters 7.1 input voltage 7.2 motor rated voltage 7.3 Motor full load current 7.4 Compressor motor 7.5 Thermostat 7.6 Cooling fan 7.7 Defrost heater 7.8 Timer 7.9 Relay 7.10 Compressor motor 7.11 Thermal fuse 7.12 Capacitors Mechanical parameter 7.13 Type of refrigerant 7.14 Type of expansion valve (refrigerant flow control) 7.15 Type of condenser, (Serpentine, compact, static, forced circulation) 7.16 Type of evaporator; (serpentine, compact, static, forced circulation)
8. System trouble/problem	8.1 Input electrical/electronic problem 8.2 Faulty Electrical/electronic circuit 8.3 Faulty compressor 8.4 Faulty refrigerant charge 8.5 System leak 8.6 Faulty mechanical system components
9. Electrical/electronic Maintenance activities	9.1 Compressor motor replacement 9.2 Electrical/electronic components repair/replacement 9.3 Re-wiring 9.4 Checking/cleaning of terminals and contactors
10. Manufacturer's instructions/specification	10.1 Vacuum pressure testing 10.2 Amount of refrigerant charge 10.3 Type of refrigerant charge 10.4 Tolerance for High side pressure 10.5 Tolerance for Low side pressure 10.6 Range of full load current 10.7 Compressor power and capacity 10.8 Pressure testing
11. Compressor operation faults	11.1 Improper power supply 11.2 Loose Connection 11.3 Broken/leaky valves 11.4 Stuck up piston 11.5 Broken crankshaft 11.6 Leaky gauge manifolds 11.7 Compressor motor problem (for hermetic type)

12. Mechanical refrigeration system fault	12.1 Leaky line tubes, compressor oil, evaporator coil 12.2 Low refrigerant charge 12.3 High refrigerant charge 12.4 Partial pressure in system 12.5 Faulty expansion valve 12.6 Dirty condenser 12.7 Dirty evaporator 12.8 Faulty refrigerator/freezer door gasket 12.9 Faulty choking
13. Servicing procedures	13.1 Installation and commissioning of new units 13.2 System checks and cleaning 13.3 Replacement of refrigerant filter/drier 13.4 Replacement of system components 13.5 Electrical/electronic troubleshooting and repair 13.6 Retrofitting 13.7 Evacuation and charging of new refrigerant 13.8 Refrigerant recovery and re-cycling 13.9 Changing of tubes and brazing 13.10 Leak repair
14. Refrigerator/freezer	14.1 Household refrigerator (frost, no frost, defrost) 14.2 Reach in freezer 14.3 Deep freezer 14.4 Ice cream maker 14.5 Ice maker 14.6 Display Chiller 14.7 Display freezer

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of Operation of refrigerator/freezer 1.2 Technical information for refrigerators and freezers 1.3 Procedure for Starting and operation of refrigerator/freezer, 1.4 Electrical/electronic circuits and components of different types of refrigerators and freezers 1.5 Electrical, and mechanical technical parameters of refrigerators and freezers 1.6 Procedure of Identifying and recording electrical system trouble/problem 1.7 Electrical/electronic maintenance activities and manufacturer's instructions/specification 1.8 Methods of checking and testing operation of electrical components 1.9 Procedure of Identifying and repairing/servicing mechanical refrigeration system fault 1.10 Servicing procedures for refrigeration system components

	1.11 Testing procedure for refrigerator/freezer acceptable operating performance 1.12 Workplace requirements of cleaning, checking tools and equipment for damaged and storing
2. Underpinning Skills	2.1 Inspecting refrigerator/freezer and identifying corresponding technical information 2.2 Starting and operating refrigerator/freezer, if possible, and observing 2.3 Observing electrical, electronic and mechanical technical parameters and recording results/findings 2.4 Identifying and recording electrical/electronic system trouble/problem 2.5 Carrying out electrical maintenance activities in accordance with manufacturer's instructions/specification 2.6 Checking and testing operation of electrical components and system for proper operation in accordance with manufacturer's instruction/specification 2.7 Identifying and repairing/servicing mechanical refrigeration system fault in accordance with manufacturer's specifications 2.8 Carrying out servicing procedures for refrigeration system components in accordance with manufacturer's instructions/ specifications 2.9 Testing refrigerator/freezer for acceptable operating performance in accordance with manufacturer's specifications 2.10 Cleaning, checking tools and equipment for damaged and lubricated (if necessary) and storing in accordance with workplace conditions.
3. Underpinning Attitudes	3.1 Patience 3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns 3.4 Eagerness to learn 3.5 Tidiness and orderliness 3.6 Respect for rights of peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Refrigeration and air conditioning equipment, tools, materials, repair and training manual. 4.3 Work instruction sheet 4.4 Refrigerator/freezer

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Inspected refrigerator/freezer and identified corresponding technical information 1.2 Started and operated refrigerator/freezer, if possible, and observed

	1.3 Checked and tested operation of electrical components and system for proper operation in accordance with manufacturer's instruction/specification 1.4 Identified and repaired/serviced mechanical refrigeration system fault in accordance with manufacturer's specifications 1.5 Carried out servicing procedures of refrigeration system components in accordance with manufacturer's instructions/specifications 1.6 Tested refrigerator/freezer for acceptable operating performance in accordance with manufacturer's specifications
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE AND MAINTAIN WINDOW TYPE AIR CONDITIONING SYSTEM**	**Nominal Duration: 30 hrs**	**Unit Code:** SICIP-LE-RAC-04-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to service and maintain window type air conditioning system. It specifically includes the tasks of carrying out preparation for servicing and maintenance works; troubleshooting window air conditioner; maintaining/repairing electrical /electronic system; servicing mechanical refrigeration system; installing window type air conditioner; cleaning /maintaining workplace, tools and equipment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out preparation for servicing and maintenance works	1.1 Safe work practices are maintained and Personal Protective Equipment (PPE) are used. 1.2 Workplace area is inspected and hazards are eliminated. 1.3 Tools, equipment & materials are gathered. 1.4 Tools, equipment & materials are checked for usability and quality. 1.5 Workplace is prepared for servicing activities. 1.6 Window type air conditioner is inspected and corresponding technical information is identified and interpreted.
2. Troubleshoot window air conditioner	2.1 Relevant information regarding trouble/problem is asked from user/owner of unit. 2.2 Electrical wiring circuit is checked and traced. 2.3 Electronic circuit is checked and traced. 2.4 Window air conditioner is started and operated, if possible, and observed operation. 2.5 **Electrical, electronic and mechanical parameters** are observed and recorded. 2.6 System trouble/problem is identified, and results/findings are recorded.
3. Maintain/repair electrical /electronic system	3.1 Electrical trouble/problem is identified. 3.2 Faulty **Electrical and electronic component/s** is tested and repaired/replaced where necessary. 3.3 Specifications of electrical/electronic component for replacement is checked and recorded. 3.4 **Electrical Maintenance activities** are carried out in accordance with **manufacturer's instructions/specification.** 3.5 Operation of electrical/electronic components and system is checked and tested for proper operation in accordance with manufacturer's specification.

4. Service mechanical refrigeration system	4.1 Compressor operation fault is identified. 4.2 Compressor fault is repaired and tested for normal operation. 4.3 Mechanical refrigeration system fault is identified and repaired/serviced in accordance with manufacturer's instructions/specifications. 4.4 **<u>Servicing procedures</u>** of window type air conditioning system components are carried out in accordance with manufacturer's instructions/specifications. 4.5 Window air conditioner is tested for acceptable operating performance in accordance with manufacturer's specifications.
5. Install window type air conditioner	5.1 Wall is cut in accordance with the specification 5.2 Outside frame is set up 5.3 Window type air conditioner is commissioned and run
6. Clean/maintain workplace, tools and equipment	6.1 Workplace is cleaned and materials are stored in accordance with workplace requirements. 6.2 Tools and equipment are cleaned, checked for damaged and lubricated (if necessary) and stored in accordance with workplace conditions. 6.3 Damaged/defective tools and equipment are reported for repair/replacement.

Range of Variables

Variable	**Range** (Includes but not limited to:)
1. System trouble/problem	1.1 Input electrical problem 1.2 Faulty Electrical circuit 1.3 Electronic circuit problems 1.4 Electronic component problem 1.5 Faulty compressor 1.6 Faulty blower fan motor 1.7 Faulty refrigerant charge 1.8 System leak 1.9 Faulty capacitor 1.10 Faulty mechanical system
2 Electrical, electronic and mechanical parameters	2.1 Electrical/ electronic parameters 2.1.1 input voltage 2.1.2 motor rated voltage 2.1.3 Motor full load current 2.1.4 Cycle 2.1.5 Motor phase; (single phase, three phase) 2.1.6 Electronic circuits 2.1.7 Resistance 2.1.8 Inductance 2.1.9 Capacitance

Variable	**Range** (Includes but not limited to:)
	2.1.10 Semiconductor devices 2.1.11 Transformer 2.2 Mechanical parameters 2.2.1 High side pressure 2.2.2 Low side pressure 2.2.3 Type of compressor; (Piston type, Rotary type) 2.2.4 Type of refrigerant 2.2.5 Type of expansion valve (refrigerant flow control) 2.2.6 Type of condenser, 2.2.7 Type of evaporator
3 Electrical/Electronic component/s	3.1 Compressor motor 3.2 Relays 3.3 Timers 3.4 Switches 3.5 Thermostat 3.6 Lamps 3.7 Timer relays 3.8 Wiring (includes terminals and connectors) 3.9 Pressure switches 3.10 Capacitors 3.11 Resistors 3.12 Diode 3.13 Inductors 3.14 Capacitors 3.15 transformers
4 Electrical/electronic Maintenance activities	4.1 Compressor motor replacement 4.2 Electrical/ electronic components repair/replacement 4.3 Re-wiring 4.4 Checking/cleaning of terminals and contactors
5 Manufacturer's instructions/specification	5.1 Pressure testing 5.2 Vacuum pressure testing 5.3 Amount of refrigerant charge 5.4 Type of refrigerant charge 5.5 Tolerance for High side pressure 5.6 Tolerance for Low side pressure 5.7 Range of full load current 5.8 Compressor power and capacity
6 Servicing procedures	6.1 Installation and commissioning of new units 6.2 System checks and cleaning 6.3 Replacement of air filter 6.4 Replacement of refrigerant filter/drier 6.5 Replacement of system components 6.6 Electrical troubleshooting and repair 6.7 Evacuation and charging of new refrigerant 6.8 Changing of tubes and brazing 6.9 Leak repair

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of Operation of window type air conditioner 1.2 Technical information for window type air conditioning systems 1.3 Procedure for Starting and operation of window air conditioner 1.4 Electrical/electronic circuits and components of window air conditioner 1.5 Electrical/electronic and mechanical technical parameters of window air conditioner 1.6 Procedure of Identifying and recording electrical system trouble/problem 1.7 Electrical/ electronic maintenance activities and manufacturer's instructions/specification 1.8 Methods of checking and testing operation of electrical components 1.9 Procedure of Identifying and repairing/servicing mechanical refrigeration system fault 1.10 Servicing procedures for window air conditioner system components 1.11 Testing procedure for window air conditioner 1.12 Acceptable operating performance 1.13 Workplace requirements of cleaning, checking tools and equipment for damaged and storing
2. Underpinning Skills	2.1 Inspecting window air conditioner and identifying corresponding technical information 2.2 Starting and operating window air conditioner, if possible, and observing 2.3 Observing electrical/electronic and mechanical technical parameters and recording results/findings 2.4 Identifying and recording electrical/electronic system trouble/problem 2.5 Carrying out electrical/electronic maintenance activities in accordance with manufacturer's instructions/specification 2.6 Checking and testing operation of electrical/electronic components and system for proper operation in accordance with manufacturer's instruction/specification 2.7 Identifying and repairing/servicing mechanical refrigeration system fault in accordance with manufacturer's specifications 2.8 Carrying out servicing procedures for window air conditioner components in accordance with manufacturer's instructions/ specifications 2.9 Testing window air conditioner for acceptable operating performance in accordance with manufacturer's specifications

	2.10 Cleaning, checking tools and equipment for damaged and lubricated (if necessary) and storing in accordance with workplace conditions.
3. Underpinning Attitudes	3.1 Patience 3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns 3.4 Eagerness to learn 3.5 Tidiness and orderliness 3.6 Respect to peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Refrigeration and air conditioning equipment, tools, materials, repair and training manual. 4.3 Work instruction sheet 4.4 Window type air conditioner

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Inspected window air conditioner and identified corresponding technical information 1.2 Started and operated window air conditioner, if possible, and observed 1.3 Observed electrical, electronic and mechanical technical parameters and recorded results/findings 1.4 Identified and recorded electrical/electronic system trouble/problem 1.5 Carried out electrical maintenance activities in accordance with manufacturer's instructions/specification 1.6 Checked and tested operation of electrical/electronic components and system for proper operation in accordance with manufacturer's instruction/specification 1.7 Identified and repaired/serviced mechanical refrigeration system fault in accordance with manufacturer's specifications 1.8 Carried out servicing procedures of window air conditioner system components in accordance with manufacturer's instructions/specifications 1.9 Tested window air conditioner for acceptable operating performance in accordance with manufacturer's specifications 1.10 Cleaned, checked tools and equipment for damaged and lubricated (if necessary) and stored in accordance with workplace conditions.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test

	2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **SERVICE, MAINTAIN AND INSTALL SPLIT AND PACKAGE TYPE AIR CONDITIONING SYSTEM**	**Nominal Duration:120 hrs**	**Unit Code:** SICIP-LE-RAC-05-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude required to service, maintain and install split and package type air conditioning system. It specifically includes the tasks of carrying out preparation for servicing and maintenance works; troubleshooting split and package type air conditioning units; maintaining/repairing electrical and electronic system; servicing mechanical system and components; installing split type air conditioner; cleaning/maintaining workplace, tools and equipment		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1.Carry out preparation for servicing and maintenance works	1.1 Safe work practices are observed, and Personal Protective Equipment (PPE) are used. 1.2 Workplace area is inspected and hazards are eliminated. 1.3 **Tools, equipment & materials** are gathered. 1.4 Tools, equipment & materials are checked for usability and quality. 1.5 Workplace is prepared for servicing activities. 1.6 Split type air conditioner is inspected and corresponding **technical information** is identified.
2. Troubleshoot split and package type air conditioning units	2.1 **Relevant information** regarding trouble/problem is asked from user/owner of unit. 2.2 Electrical wiring circuit is checked and traced. 2.3 Split and package type air conditioning units are started and operated, if possible, and observed operation. 2.4 **Electrical/electronic and mechanical parameters** are observed and recorded. 2.5 System trouble/problem is identified and results/findings are recorded.
3. Maintain/repair electrical and electronic system	3.1 Electrical trouble/problem is identified 3.2 Faulty **Electrical/electronic components** is tested and repaired/replaced where necessary. 3.3 Inverter Circuit is checked and repaired/replaced. 3.4 Specifications of electrical component for replacement is checked and recorded. 3.5 **Electrical/electronic Maintenance activities** are carried out in accordance with **manufacturer's instructions/specification.** 3.6 Operation of electrical components and system is checked and tested for proper operation in accordance with manufacturer's specification.

4. Service mechanical system and components	4.1 **<u>Compressor operation fault</u>** is identified and interpreted. 4.2 Compressor fault is repaired and tested for normal operation. 4.3 **<u>Mechanical refrigeration system fault</u>** is identified and repaired/serviced in accordance with manufacturer's instructions/specifications. 4.4 **<u>Servicing procedures</u>** of split type air conditioning system components are carried out in accordance with manufacturer's instructions/specifications. 4.5 **<u>Split and package type air conditioning</u>** units are tested for acceptable operating performance in accordance with manufacturer's specifications.
5. Install split type air conditioner	5.1 Baseplate set up is performed in accordance with the appropriate measurement 5.2 Outdoor frame is fixed and set up 5.3 Indoor and outdoor copper tube connection is carried out 5.4 Leak and pressure test of copper tube is preformed and vacuumed 5.5 Refrigerant charging is carried out as the standard procedure 5.6 Split type air conditioner is commissioned and run
6. Clean/maintain workplace, tools and equipment	6.1 Workplace is cleaned and materials are stored in accordance with workplace requirements. 6.2 Tools and equipment are cleaned, checked for damaged and lubricated (if necessary) and stored in accordance with workplace conditions. 6.3 Damaged/defective tools and equipment are reported for repair/replacement.

Range of Variables

Variable	**Range** (Includes but not limited to)	
1. Tools, equipment and materials	1.1 **Tools:** 1.1.1 Multi tester (AVO) 1.1.2 Clamp meter 1.1.3 Side cutting pliers 1.1.4 Wire stripper 1.1.5 Screwdriver set 1.1.6 Adjustable wrench 1.1.7 Allen Key set 1.1.8 Chisel 1.1.9 Terminal crimping plier 1.1.10 Measuring steel tape 1.1.11 Ball peen hammer	1.2.3 Refrigerant recovery machine 1.2.4 Oxy acetylene welding outfit 1.2.5 Power Drill machine 1.2.6 Double gauge manifold 1.2.7 Nitrogen Cylinder 1.2.8 Double stage regulator 1.3 **Materials:** 1.3.1 Insulation tapes 1.3.2 Electrical terminal connectors

	1.1.12 Tube cutter 1.1.13 Hand hacksaw 1.1.14 Swaging tool set 1.1.15 Flaring tool set 1.1.16 Bending tool 1.1.17 Files 1.2 **Equipment:** 1.2.1 Vacuum pump 1.2.2 Refrigerant charging cylinder	1.3.3 Wires/cables 1.3.4 Copper tubes 1.3.5 Aluminum tubes 1.3.6 Steel tube 1.3.7 Brazing flux 1.3.8 Brazing Rod 1.3.9 Insulating tube 1.3.10 MS Angle/Flat bar
2. Technical information	2.1 Type of refrigerant used 2.2 High side systems pressure 2.3 Low side system pressure 2.4 Type of compressor used 2.5 Voltage of compressor motor 2.6 Full load current rating 2.7 Cycle/second (Hz) 2.8 Electrical/electronic circuit diagram 2.9 Electrical/electronic components 2.10 Type of evaporator coil 2.11 Type of condensing unit 2.12 Length of copper tubes	
3. Relevant information	3.1 Power supply 3.2 Electrical/electronic circuit 3.3 System operation 3.3.1 Condensing unit 3.3.2 Compressor 3.3.3 Evaporator 3.3.4 Condenser 3.3.5 Expansion valve 3.3.6 Refrigerant charge 3.3.7 Leaks 3.4 Incidents prior to occurrence of problem	
4. Electrical, electronic and mechanical parameters	4.1 Electrical parameters 4.1.1 input voltage 4.1.2 motor rated voltage 4.1.3 Motor full load current 4.1.4 Cycle/sec(Hz) 4.1.5 Motor phase; (single phase, three phase) 4.2 Mechanical parameters 4.2.1 High side pressure 4.2.2 Low side pressure 4.2.3 Type of compressor; (Piston type, Rotary type) 4.2.4 Type of refrigerant 4.2.5 Type of expansion valve (refrigerant flow control) 4.2.6 Type of condensing unit 4.2.7 Type of evaporator (indoor unit)	

	4.3 Electronic parameters 4.3.1 Electronic circuits 4.3.2 Resistance 4.3.3 Inductance 4.3.4 Capacitance 4.3.5 Semiconductor devices 4.3.6 Transformer
5. System trouble/problem	5.1 Input electrical problem 5.2 Faulty Electrical circuit 5.3 Faulty compressor 5.4 Faulty refrigerant charge 5.5 System leak 5.6 Faulty mechanical system components
6. Electrical and electronic component/s	6.1 Compressor motor 6.2 Contactor 6.3 Relays 6.4 Timers 6.5 Switches 6.6 Thermostat 6.7 Lamps 6.8 Timer relays 6.9 Wiring (includes terminals and connectors) 6.10Pressure switches 6.11Capacitors 6.12Resistor 6.13Diode. 6.14Capacitor 6.15Inductor 6.16Transistor 6.17 IC
7. Electrical and electronic Maintenance activities	7.1 Compressor motor replacement 7.2 Electrical and electronic components repair/replacement 7.3 Re-wiring 7.4 Checking/cleaning of terminals and contactors
8. Manufacturer's instructions/specification	8.1 Pressure testing 8.2 Vacuum pressure testing 8.3 Amount of refrigerant charge 8.4 Type of refrigerant charge 8.5 Tolerance for High side pressure 8.6 Tolerance for Low side pressure 8.7 Range of full load current 8.8 Compressor power and capacity
9. Compressor operation fault	9.1 Loose connection 9.2 Broken/leaky valves 9.3 Stuck up piston

	9.4 Broken crankshaft 9.5 Leaky manifolds 9.6 Compressor motor problem (for hermetic type)
10. Mechanical refrigeration system fault	10.1 Leaky line tubes, compressor oil, evaporator coil 10.2 Low refrigerant charge 10.3 High refrigerant charge 10.4 Partial pressure in system 10.5 Faulty expansion valve 10.6 Dirty condenser 10.7 Dirty evaporator 10.8 Faulty refrigerator/freezer door sealant
11. Servicing procedures	11.1 Installation of new units 11.2 Indoor and outdoor units checks and cleaning 11.3 Replacement of air filter 11.4 Replacement of refrigerant filter/drier 11.5 Replacement of system components 11.6 Electrical troubleshooting and repair 11.7 Retrofitting 11.8 Evacuation and charging of new refrigerant 11.9 Changing of tubes and brazing 11.10 Leak repair
12. Split and package type air conditioning	12.1 Split type air conditioning units 12.1.1 Floor mounted indoor unit 12.1.2 Wall mounted indoor unit 12.1.3 Ceiling mounted indoor unit 12.1.4 Portable type unit 12.2 Package type air conditioning unit 12.2.1 floor mounted compact type 12.2.2 single ducted type 12.2.3 multiple ducted type 12.2.4 Variable Refrigerant Flow(VRF)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Refrigerant use and types of refrigerant 1.2 Principles of Operation of split type air conditioning systems 1.3 Principles of operation of package type air conditioning units 1.4 Types of split type air conditioning systems 1.5 Technical information for split type air conditioning systems 1.6 Technical information for package type air conditioning systems 1.7 Procedure for Starting and operation of split and package type conditioner 1.8 Electrical/electronic circuits and components of split and package type air conditioners

	1.9 Electrical, electronic and mechanical technical parameters of split and package type air conditioners 1.10 Procedure of Identifying and recording electrical system trouble/problem 1.11 Electrical/electronic maintenance activities and manufacturer's instructions/specification 1.12 Methods of checking and testing operation of electrical components 1.13 Procedure of Identifying and repairing/servicing mechanical refrigeration system fault 1.14 Servicing procedures for split and package air conditioner system components 1.15 Testing procedure for split and package air conditioner 1.16 Acceptable operating performance 1.17 Workplace requirements of cleaning, checking tools and equipment for damaged and storing
2. Underpinning Skills	2.1 Inspecting split and package air conditioner and identifying corresponding technical information 2.2 Starting and operating split and package air conditioner, if possible, and observing 2.3 Observing electrical, electronic and mechanical technical parameters and recording results/findings 2.4 Identifying and recording electrical/electronic system trouble/problem 2.5 Carrying out electrical/electronic maintenance activities in accordance with manufacturer's instructions/specification 2.6 Checking and testing operation of electrical/electronic components and system for proper operation in accordance with manufacturer's instruction/specification 2.7 Identifying and repairing/servicing mechanical refrigeration system fault in accordance with manufacturer's specifications 2.8 Carrying out servicing procedures for split and package air conditioner components in accordance with manufacturer's instructions/ specifications 2.9 Testing split and package air conditioners for acceptable operating performance in accordance with manufacturer's specifications 2.10 Cleaning, checking tools and equipment for damaged and lubricated (if necessary) and storing in accordance with workplace conditions.
3. Underpinning Attitudes	3.1 Patience 3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns

	3.4 Eagerness to learn 3.5 Tidiness and orderliness 3.6 Respect for rights of peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Refrigeration and air conditioning equipment, tools, materials, repair and training manual. 4.3 Work instruction sheet 4.4 Split and package type air conditioning manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Inspected split and package type air conditioner and identified corresponding technical information 1.2 Started and operated split and package type air conditioner, if possible, and observed 1.3 Observed electrical, electronic and mechanical technical parameters and recorded results/findings 1.4 Identified and recorded electrical/electronic system trouble/problem 1.5 Carried out electrical maintenance activities in accordance with manufacturer's instructions/specification 1.6 Checked and tested operation of electrical/electronic components and system for proper operation in accordance with manufacturer's instruction/specification 1.7 Carried out servicing procedures of split and package type air conditioner system components in accordance with manufacturer's instructions/specifications 1.8 Tested split and package type air conditioner for acceptable operating performance in accordance with manufacturer's specifications 1.9 Cleaned, checked tools and equipment for damaged and lubricated (if necessary) and stored in accordance with workplace conditions.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT COMPRESSOR SERVICING**	**Nominal Duration: 20 hrs**	**Unit Code:** SICIP-LE-RAC-06-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to carry out compressor servicing. It specifically includes the tasks of carrying out preparation for compressor servicing/repairing; diagnosing fault and servicing compressor; cleaning/maintaining workplace, tools and equipment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out preparation for compressor servicing/repairing	1.1 Safe work practices are maintained and Personal Protective Equipment (PPEs) are used. 1.2 Workplace area is inspected and hazards are eliminated. 1.3 Tools, equipment & materials are gathered and interpreted. 1.4 Tools, equipment & materials are checked for usability and quality. 1.5 Workplace is prepared for repairing of refrigeration compressor. 1.6 **Compressor technical information** is identified interpreted and recorded.
2. Diagnose fault and service compressor	2.1 **Type of faulty compressor** is identified. 2.2 **Relevant information** regarding trouble/problem is asked from user/owner of unit. 2.3 Refrigerant is recovered appropriately 2.4 Compressor is removed from refrigeration unit in accordance with workplace/manufacturer's instruction. 2.5 Compressor is disassembled in accordance with manufacturer's instruction/specification. 2.6 **Compressor parts/components** are checked and faults identified 2.7 Compressor is serviced or replaced
3. Clean/maintain workplace, tools and equipment	3.1 Workplace is cleaned and materials are stored in accordance with workplace requirements. 3.2 Tools and equipment are cleaned, checked for damaged and lubricated (if necessary) and stored in accordance with workplace conditions. 3.3 Damaged/defective tools and equipment are reported for repair/replacement.

Range of Variables

Variable	**Range** (Includes but not limited to:)

1. Compressor technical information	1.1 Type of compressor 1.2 Types of Suction and discharge valves 1.3 Manufacturer's Serial number 1.4 Manufacturer's part numbers 1.5 Type of refrigerant used 1.6 Working pressure 1.7 Electrical specifications 1.8 Compressor oil
2. Type of compressor	2.1 Reciprocating Compressors. 2.2 Rotary type Compressors. 2.3 Scroll Compressors. 2.4 Centrifugal Compressors.
3. Relevant information	3.1 Time when trouble was detected 3.2 Abnormal noise 3.3 Power supply condition 3.4 Abnormal temperature 3.5 Abnormal pressures 3.6 Other abnormalities of operation observed 3.7 Incidents prior to occurrence of problem
4. Compressor parts/components	4.1 For reciprocating compressor 4.1.1 Cylinder head 4.1.2 Suction valves 4.1.3 Valve plate 4.1.4 Discharge valve 4.1.5 Piston 4.1.6 Connecting rod 4.1.7 Crankshaft 4.1.8 Bearings 4.1.9 Seals 4.1.10 Fly wheel 4.2 For rotary compressor 4.2.1 Suction port 4.2.2 Discharge port 4.2.3 Pressure regulating valve 4.2.4 Rotor 4.2.5 Cylinder 4.2.6 Shaft 4.2.7 Coupling 4.2.8 Seals 4.2.9 Check valves

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of compressors and their applications 1.2 Compressor testing procedure 1.3 Procedure of removing compressor from refrigeration system 1.4 Compressor disassembly procedure

	1.5 Compressor parts/components checking and fault analysis 1.6 Parts and components of compressors and their functions 1.7 Parts/components repair procedures 1.8 Compressor Testing procedure 1.9 Manufacturer's specifications for different types of compressors 1.10 Workplace requirements for tools and equipment Maintenance and storing in requirements
2. Underpinning Skills	2.1 Testing compressor, if possible, and observing operation 2.2 Removing compressor from refrigeration system in accordance with workplace/manufacturer's instruction 2.3 Disassembling compressor in accordance with manufacturer's instruction/specification 2.4 Checking compressor parts/components and identifying faults 2.5 Repairing faulty parts/components or replacing where necessary in accordance with workplace/manufacturer's instruction/specification 2.6 Assembling compressor parts/components in accordance with manufacturer's instruction/specification 2.7 If nor repaired, compressor is replaced 2.8 Testing compressor for normal operation in accordance with manufacturer's specifications 2.9 Maintaining tools and equipment and storing in accordance with workplace requirements
3. Underpinning Attitudes	3.1 Patience 3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns 3.4 Eagerness to learn 3.5 Tidiness and orderliness 3.6 Respect for rights of peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Refrigeration and air conditioning equipment, tools, materials, repair and training manual. 4.3 Work instruction sheet 4.4 Refrigerant compressor

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Tested compressor, if possible, and observed operation

	1.2 Removed compressor from refrigeration system in accordance with workplace/manufacturer's instruction 1.3 Disassembled compressor in accordance with manufacturer's instruction/specification 1.4 Checked compressor parts/components and identified faults 1.5 Repaired faulty parts/components or replaced where necessary in accordance with workplace/manufacturer's instruction/specification 1.6 assembled compressor parts/components in accordance with manufacturer's instruction/specification 1.7 Replaced compressor if required.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	**Refrigeration and Air Conditioning**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom and Workshop/ Lab):

- Classroom : 350 sft
- Workshop/Lab : 800-1000 sft

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Split Type AC (1 Inverter type and 4 non-inverter type)	5
2.	Window Type AC	2
3.	Cassette Type AC	1
4.	Vacuum Pumps	5
5.	Swaging Tools set	5
6.	Refrigeration Model	2
7.	Flaring Tools set	5
8.	Gas welding Set and its accessories (Cylinder with O_2 and acetylene gas)	1
9.	Working Table	5
10.	Leak detectors	2
11.	Freeze (Frost)	5
12.	Freeze (Non-Frost)	2
13.	Hand Grinder	2
14.	Cylinder with Nitrogen gas	2
15.	Electric Hand Drill Machine	2
16.	Hammer Drill Machine	2
17.	Refrigerant Recovery Machine	2
18.	Refrigerant Recovery Cylinder	2

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Refrigeration and Air Conditioning, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.